BEmoMix: Multi-Label Banglish Emotion Detection Using Transformer and Recurrent Models

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Abstract

Emotion detection in Bangla–English code-mixed (Banglish) text is crucial for understanding user affect in social media, yet existing resources primarily target native-script Bengali or sentiment polarity with single-label assumptions. This paper introduces BEmoMix, a large-scale Banglish emotion dataset of 27,809 Bangla texts written in Roman script, annotated with seven emotion categories (Joy, Sadness, Anger, Fear, Hate, Surprise, Love) under a multi-label scheme that allows multiple emotions per instance. A unified modeling framework was developed to evaluate a diverse set of baseline and transformer-based architectures for multi-label Banglish emotion detection. The evaluated models include BiLSTM, BiLSTM with Attention, Multilingual-e5-base, Ensemble (e5-base), multilingual BERT (mBERT), XLM-R Base, XLM-R Large variants, mDeBERTa-based hybrids, and the proposed XLMR-OPT architecture. Experimental findings demonstrate that transformer-based multilingual models significantly outperform recurrent neural and embedding-based baselines across all evaluation metrics. Among all approaches, the proposed XLMR-OPT model achieved the best overall performance with a Macro-F1 score of 0.7495, Hamming Accuracy of 0.9154, and Strict Accuracy of 0.6240, highlighting its effectiveness for capturing complex multi-label emotional patterns in Banglish code-mixed text. BEmoMix establishes strong baselines for Banglish emotion detection and opens new directions for label-correlation–aware modeling and cross-lingual transfer in low-resource, code-mixed settings.

CCS Concepts

• **Computing methodologies → Natural language processing**; *Machine learning*; *Neural networks.*

Keywords

Banglish, emotion detection, multi-label classification, code-mixed text, transformer models, multilingual NLP

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1 Introduction

Emotion detection from textual data has become an important task in natural language processing (NLP), enabling computational systems to identify human affective states from digital communication [1, 10]. Emotion-aware NLP applications are increasingly used in mental health monitoring, recommendation systems, content moderation, and social media analytics. However, emotion recognition in low-resource and code-mixed languages remains challenging.

One prominent example is *Banglish*, where Bangla is written using Roman script and frequently mixed with English expressions [3, 6]. Banglish communication is highly common on social media platforms in Bangladesh and is characterized by inconsistent transliteration, phonetic spelling variation, informal abbreviations, and frequent code-switching. These properties increase lexical variability and reduce the effectiveness of conventional NLP systems.

Existing studies on Bengali emotion analysis primarily focus on native-script Bengali text or coarse-grained sentiment classification [1, 8]. Most prior approaches also assume a single-label setting, where each text is associated with only one dominant emotion. In practice, however, social media expressions often contain multiple overlapping emotional states, making single-label modeling insufficient for realistic emotion understanding.

Although several datasets and models have been proposed for Bengali emotion and code-mixed sentiment analysis, large-scale resources specifically designed for multi-label Banglish emotion detection remain limited [9, 15]. Recent transformer-based approaches have shown promising performance for Bengali emotion classification [2, 5], yet systematic evaluation in noisy Romanized Banglish environments is still underexplored.

To address these limitations, this paper introduces **BEmoMix**, a large-scale Banglish dataset containing 27,809 Romanized social media texts annotated across seven emotion categories: Joy, Sadness, Anger, Fear, Hate, Surprise, and Love. In addition, this work proposes **XLMR-OPT**, an optimized multilingual transformer framework for robust multi-label emotion classification in code-mixed environments.

The main contributions of this work are summarized as follows:

- **Large-scale multi-label Banglish dataset:** Introduction of BEmoMix, a dataset of 27,809 Romanized Bangla texts annotated with seven emotion categories.
- **Comprehensive benchmarking:** Evaluation of multiple recurrent and transformer-based architectures under a unified experimental setting.
- **Proposed XLMR-OPT framework:** An optimized multilingual transformer architecture incorporating hybrid contextual pooling, focal optimization, and adaptive threshold calibration.
- **Extensive analysis:** Quantitative and qualitative analyses of label correlation, model behavior, and common prediction challenges in Banglish emotion detection.

2 Literature Review

2.1 Single-Label Emotion Detection in Bangla

Early research on Bengali emotion recognition primarily focused on monolingual, native-script texts with single-label annotations. Das et al. introduced *EmoNoBa*, a Banglish emotion dataset with fine-grained emotion categories and baseline models such as BiLSTM and CNN architectures [9]. Although this work established an important benchmark, it did not address code-mixed Banglish or multi-label emotion classification.

Further studies explored emotion and sentiment detection in Bengali using deep learning approaches. Biswas et al. investigated speech-based emotion classification, demonstrating the effectiveness of neural models in affect recognition tasks [10]. Chowdhury et al. analyzed emotion patterns in Bangladeshi social media data during COVID-19, highlighting the importance of contextual emotion understanding [1]. However, these works remained limited to single-label or sentiment-focused tasks.

Recent transformer-based studies by Sarkar and Saha showed that pretrained multilingual models significantly improve emotion classification performance in Bengali NLP tasks [5]. However, multi-label Banglish emotion detection remains unexplored.

2.2 Code-Mixed and Banglish Sentiment Analysis

With the increasing prevalence of code-mixed communication, sentiment analysis in Banglish has gained attention. Neogi et al. introduced a deep learning-based sentiment analysis framework for Banglish documents, showing strong performance over classical methods [6]. Similarly, Akter et al. applied machine learning and deep learning techniques for Banglish sentiment classification [3].

Alam et al. proposed *BnSentMix*, a Bengali-English code-mixed sentiment dataset, and demonstrated the effectiveness of transformer-based models for sentiment classification tasks [4]. However, all these works focus on sentiment polarity rather than fine-grained emotion categories.

2.3 Multilingual Transformers and Recent Advances

The rise of multilingual transformer models has significantly improved performance in low-resource NLP tasks. Raihan et al. introduced *Mixed-Distil-BERT*, a code-mixed language model trained on multilingual data, showing strong performance in sentiment and emotion tasks [16]. This highlights the importance of adapting pretraining to code-mixed languages.

Raihan et al. proposed *EmoMix-3L*, a tri-lingual dataset for emotion detection, demonstrating the effectiveness of multilingual transformer architectures in cross-lingual emotion modeling [17].

Recent work by Rumy et al. proposed a transformer-based multi-label emotion detection framework for Bengali text, confirming the importance of multi-label modeling in emotion recognition tasks [18]. Additionally, Ahmed et al. demonstrated the effectiveness of contextual transformer models for Bengali emotion recognition [2].

Kabir et al. introduced *BEmoLexBERT*, a hybrid model for multi-label emotion classification in Bengali, further validating transformer-based approaches in emotion detection tasks [11].

2.4 Motivation and Research Gap

Despite significant progress in Bengali emotion detection, code-mixed sentiment analysis, and multilingual transformer models, a key limitation remains: the absence of a large-scale Banglish-specific multi-label emotion dataset.

Existing works either focus on single-label classification, sentiment polarity instead of emotion categories, or native Bengali script instead of Romanized Banglish. Furthermore, systematic evaluation of transformer-based architectures in multi-label Banglish emotion detection is still lacking.

To address these limitations, this paper introduces **BEmoMix**, a large-scale multi-label Banglish emotion dataset along with a comprehensive benchmarking framework.

3 BEmoMix Dataset

This section describes the construction, annotation, and characteristics of the proposed BEmoMix dataset, a large-scale resource designed for multi-label emotion detection in Banglish text.

3.1 Data Collection

The BEmoMix dataset comprises a total of **27,809 Banglish sentences**, written in Romanized Bangla. The dataset is constructed from two primary sources:

- **Pre-existing data:** 15,648 instances obtained from the BNSENTMIX dataset [4], originally developed for single-label sentiment classification.
- **Newly collected data:** 12,161 samples manually scraped from social media platforms, online forums, and community discussions.

All collected texts reflect real-world user-generated content and exhibit natural code-mixing between Bangla and English, consistent with prior studies on Banglish and code-mixed NLP [6, 15].

3.2 Annotation Scheme

Each instance in the dataset is annotated using a **multi-label emotion classification framework**, allowing multiple emotions to be assigned to a single text. The annotation includes seven emotion categories:

- Joy
- Sadness
- Anger
- Fear
- Hate
- Surprise
- Love

This multi-label approach aligns with recent work emphasizing the importance of capturing overlapping emotional states in real-world text [18].

3.3 Annotation Process

The dataset was manually annotated to ensure semantically consistent multi-label emotion labeling. Prior to annotation, all texts were normalized to lowercase to reduce lexical variability caused by inconsistent Romanized spelling.

Three annotators with native-level Bangla proficiency and familiarity with Banglish social media discourse independently assigned one or more emotion labels from seven categories: Joy, Sadness, Anger, Fear, Hate, Surprise, and Love. Annotation guidelines were developed to maintain consistency, including definitions, representative examples, and handling strategies for sarcasm, implicit sentiment, and overlapping emotions, following practices in recent multilingual emotion detection studies [2, 18].

To assess annotation reliability, Fleiss' Kappa coefficient was computed, achieving a score of 0.81, indicating substantial agreement among annotators. Disagreements were resolved through majority voting and consensus-based discussion sessions.

This annotation framework preserves the naturally complex and multi-label characteristics of Banglish social media emotion expression.

3.4 Dataset Characteristics

The BEmoMix dataset exhibits several challenging characteristics:

- Code-mixed language usage (Bangla + English)
- Non-standard spelling due to phonetic transliteration
- Informal expressions and abbreviations
- Multi-label emotional overlap within single instances

These properties make BEmoMix a realistic and challenging benchmark, consistent with challenges reported in code-mixed NLP research [16, 17].

3.5 Summary

BEmoMix provides a diverse and large-scale benchmark for Banglish emotion detection, addressing key limitations of prior work by supporting multi-label classification and focusing on Romanized Bangla text. It complements existing datasets such as EmoNoBa [9] while extending to a multi-label setting.

4 Methodology

This section presents the proposed XLMR-OPT framework for multi-label emotion classification in multilingual and code-mixed Banglish text. Although pretrained multilingual transformers provide strong contextual representations, directly applying standard XLM-RoBERTa [7] to Banglish emotion detection remains insufficient due to inconsistent Romanized spelling, noisy social media expressions, overlapping emotional semantics, and imbalanced multi-label co-occurrence patterns. To address these challenges, the proposed framework integrates imbalance-aware optimization, hybrid contextual feature aggregation, and adaptive threshold calibration within a unified multi-label learning architecture.

4.1 Overview of the Proposed Framework

The proposed XLMR-OPT pipeline consists of six major stages:

(1) Data preprocessing and targeted augmentation
(2) Subword tokenization using XLM-R tokenizer
(3) Contextual representation learning using pretrained XLM-RoBERTa [7]
(4) Hybrid feature extraction using CLS embedding and attention pooling
(5) Imbalance-aware optimization using focal loss [12] with label smoothing [19]
(6) Adaptive per-class threshold calibration for multi-label prediction

Unlike conventional transformer fine-tuning approaches, the proposed framework jointly integrates contextual attention aggregation, focal optimization, and threshold adaptation to improve prediction robustness in low-resource Banglish emotion detection settings.

4.2 Dataset and Preprocessing

The proposed framework is trained on the BEmoMix dataset, which consists of 27,809 Banglish social media texts annotated with seven emotion categories: Love, Joy, Anger, Surprise, Sadness, Fear, and Hate. Since multiple emotions may coexist within a single text instance, the task is formulated as a multi-label classification problem.

The dataset is divided into training, validation, and testing subsets using a stratified split strategy with a ratio of 70:15:15. A fixed random seed is used to ensure reproducibility and consistent label distribution across all partitions.

Although the individual emotion categories remain relatively balanced, the distribution of multi-label emotion co-occurrence patterns is considerably uneven. Frequent emotional combinations dominate the training distribution, whereas rare label combinations occur sparsely, creating additional challenges for learning inter-label dependencies.

Before training, all texts are normalized through lowercasing and whitespace standardization. Since Banglish social media text contains irregular transliteration, phonetic spelling variation, abbreviations, and code-switching behavior, preprocessing helps reduce lexical variability while preserving contextual meaning.

4.3 Model Architecture

The overall architecture of the proposed XLMR-OPT framework is illustrated in Fig. 1.

4.3.1 Targeted Data Augmentation. To improve representation robustness for sparse emotional combinations, targeted augmentation is applied using synonym replacement based on WordNet [14]. Selected words are replaced with semantically similar alternatives using a replacement probability of 0.2 while preserving sentence-level meaning.

This augmentation strategy increases lexical diversity and improves model generalization without introducing excessive semantic distortion.

4.3.2 Input Representation and Tokenization. Input text is tokenized using the pretrained XLM-R tokenizer. Subword tokenization is particularly effective for Banglish because code-mixed text frequently contains inconsistent transliteration and hybrid multilingual expressions. The tokenizer configuration is defined as follows:

- Maximum sequence length: 192

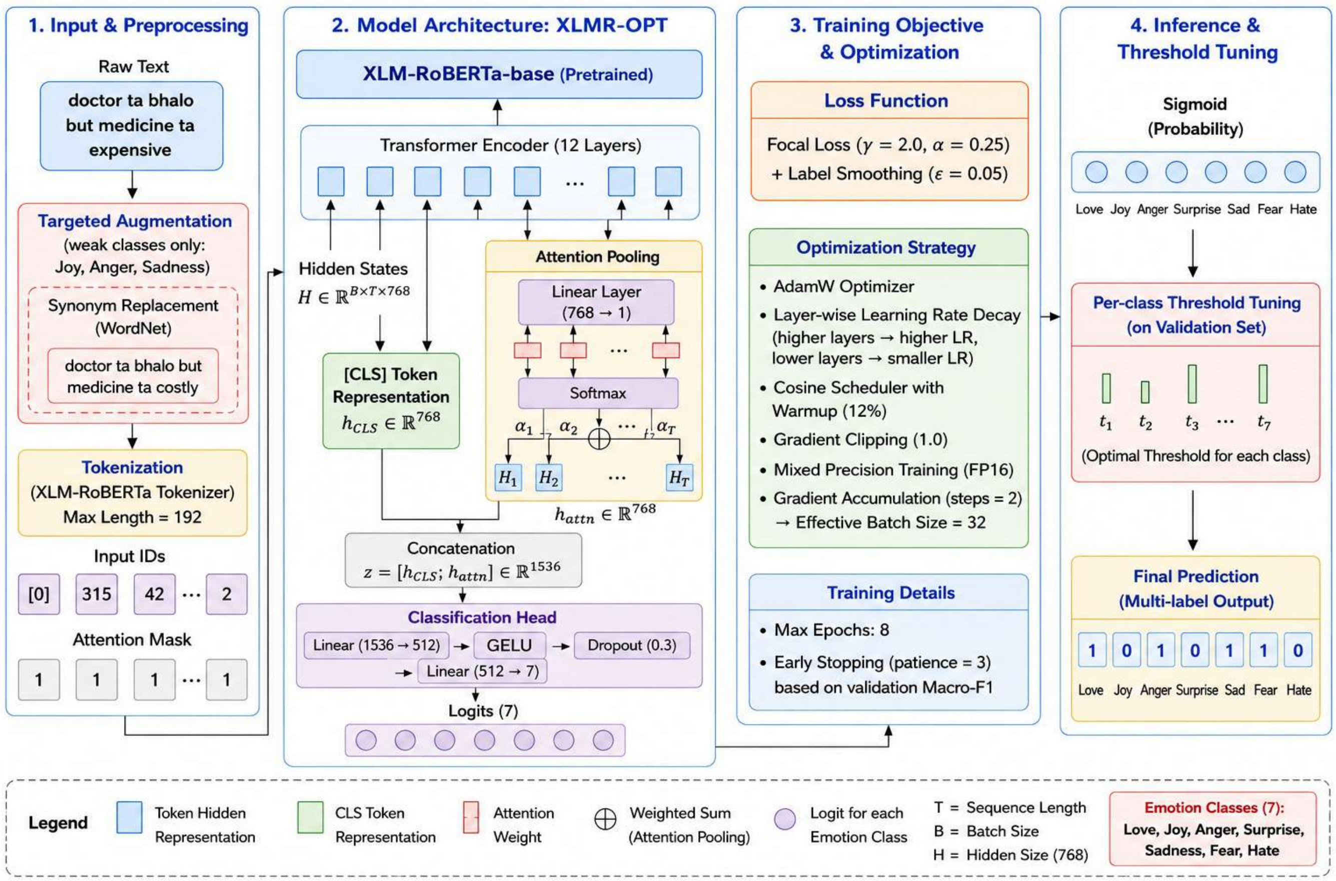

Figure 1: Overall architecture of the proposed XLMR-OPT framework for multi-label Banglish emotion classification.

- Padding: Enabled
- Truncation: Enabled

Each input sentence is converted into token IDs and attention masks before being passed into the transformer encoder.

4.3.3 Transformer-Based Contextual Representation. The proposed framework utilizes XLM-RoBERTa-base [7] as the primary contextual encoder. Given an input sequence of length T, the transformer encoder generates contextual hidden representations $H \in \mathbb{R}^{T \times 768}$, where T denotes sequence length and 768 represents the hidden embedding dimension. The standard CLS embedding captures global sentence-level semantics but may overlook localized emotional cues distributed across multiple tokens. To preserve both global contextual information and token-level emotional salience, the proposed framework combines CLS representation with attention-based pooling over transformer hidden states.

4.3.4 Hybrid CLS and Attention Pooling. Attention pooling dynamically identifies emotionally important contextual tokens. The attention-based contextual representation is computed as:

$$c = \sum_{i=1}^{n} \alpha_i h_i \tag{1}$$

where h_i represents the hidden representation of token i and α_i denotes the learned attention weight.

The final sentence representation is obtained through concatenation:

$$h_{\text{final}} = [h_{\text{CLS}}; c] \tag{2}$$

where h_{CLS} denotes the transformer-based global sentence representation and c represents the attention-pooled contextual vector.

This hybrid representation preserves both sentence-level semantics and localized emotional context, improving robustness in multi-label Banglish emotion classification.

The fused representation is passed through a feed-forward classification head consisting of linear projection, GELU activation, dropout regularization, and a final sigmoid output layer.

4.3.5 Loss Function. To address imbalance in multi-label co-occurrence patterns and improve learning from difficult samples, focal loss [12] with label smoothing [19] is employed:

$$\mathcal{L}_{\text{focal}} = -\alpha(1 - p_t)^{\gamma} \log(p_t) \tag{3}$$

where p_t denotes the smoothed target probability for the corresponding class. The hyperparameters $\gamma = 2.0$ and $\alpha = 0.25$ are used to emphasize hard samples and reduce class imbalance during optimization. Focal loss dynamically down-weights easy examples and prioritizes difficult samples during optimization. Label smoothing with a smoothing factor of 0.05 is additionally applied to reduce overconfidence and improve generalization.

4.3.6 Optimization Strategy. The model is optimized using the AdamW optimizer [13] with layer-wise learning rate decay (LLRD). Higher transformer layers are assigned larger learning rates, whereas lower layers receive smaller rates to preserve pretrained multilingual knowledge.

The training configuration includes:

- Base learning rate: 2×10^{-5}
- Cosine learning-rate scheduler with 12% warmup
- Batch size: 16
- Effective batch size: 32 via gradient accumulation
- Maximum epochs: 8
- Gradient clipping: 1.0
- Mixed precision training (FP16)
- Early stopping with patience = 3

4.3.7 Adaptive Threshold Optimization. In multi-label emotion classification, different emotion categories exhibit distinct probability distributions and occurrence frequencies. Therefore, adaptive per-class threshold tuning is performed on the validation set.

For each emotion category c, the optimal threshold t_c is selected by maximizing validation F1-score, defined as $t_c = \arg\max F1(y, \hat{y}(t))$. This adaptive calibration strategy improves class-wise prediction balance and enhances sensitivity toward sparse multi-label combinations.

4.4 Evaluation Metrics

The proposed framework is evaluated using both label-level and instance-level metrics:

- Macro Precision
- Macro Recall
- Macro F1-score
- Hamming Accuracy
- Exact Match Ratio
- Jaccard Score

Macro F1-score is selected as the primary evaluation metric because it equally weights all emotion categories and prevents performance inflation caused by dominant label patterns.

4.5 Summary of the Proposed Method

The proposed XLMR-OPT framework integrates several complementary components within a unified multilingual emotion classification architecture, including hybrid CLS-attention pooling, focal loss with label smoothing, targeted augmentation, layer-wise learning rate decay, and adaptive per-class threshold optimization.

Unlike conventional transformer fine-tuning pipelines, the proposed framework is specifically designed for low-resource Banglish emotion detection, addressing challenges such as noisy transliteration, code-mixed structure, overlapping emotions, and imbalanced multi-label distributions. These components collectively improve robustness and generalization in multilingual social media emotion classification.

4.6 Experimental Setup and Evaluation Metrics

All models are evaluated under identical data splits and preprocessing conditions to ensure fair comparison. The proposed XLMR-OPT model is built on the XLM-RoBERTa base and fine-tuned for multi-label emotion classification.

Performance is evaluated using Macro F1-score (primary metric), Hamming Accuracy, and Exact Match (Strict Accuracy). Macro F1 is emphasized as it provides balanced evaluation across all emotion classes.

4.7 Comparative Performance Analysis

Table 1 and Table 2 present the comparative performance of baseline and transformer-based models.

Table 1: Performance of Baseline Models

Model	Macro F1	Hamming Acc.	Strict Acc.
Multilingual-e5-base	0.6825	0.8939	0.5590
XLMR-baseline	0.6988	0.8959	0.5877
mBERT	0.6678	0.8945	0.5661
BiLSTM	0.6372	0.8903	0.5554
BiLSTM + Attention	0.6374	0.6374	0.5563
Ensemble (e5-base)	0.6700	0.8865	0.4873

Table 2: Performance of Transformer-based Models

Model	Macro F1	Hamming Acc.	Strict Acc.
XLMR-OPT (Proposed)	**0.7495**	**0.9154**	0.6240
XLM-R + mDeBERTa + Large	0.7116	0.9055	0.5698
XLM-R + Attention Pooling	0.6688	0.8864	0.5194
XLM-R Large (192)	0.7105	0.9109	**0.6457**
XLM-R Large (160)	0.6767	0.8853	0.4802
mDeBERTa + XLM-R Base	0.6946	0.8979	0.5403
XLM-R Base	0.6674	0.8890	0.5460
mDeBERTa + REMBERT + XLM-R	0.6716	0.8884	0.4672

The proposed XLMR-OPT achieves the highest Macro F1-score (0.7495) and Hamming Accuracy (0.9154), demonstrating superior balance across emotion categories. Although XLM-R Large (192) obtains slightly higher Strict Accuracy (0.6457), its lower Macro F1 indicates weaker label-wise consistency.

Since Strict Accuracy requires an exact match of all labels, the proposed framework prioritizes balanced multi-label prediction and minority emotion detection, improving overall Macro F1 while introducing slight variations in exact label combinations. In contrast, XLM-R Large (192) produces more conservative predictions, resulting in higher exact-match accuracy but lower overall label-wise balance.

4.8 Per-Emotion Performance Analysis

Table 3 presents the class-wise performance of the proposed XLMR-OPT framework across individual emotion categories. The model achieves the highest F1-scores for Surprise (0.8084) and Fear (0.8039), indicating strong contextual understanding of explicit emotional expressions. In contrast, Hate remains comparatively challenging due to implicit semantics, sarcasm, and overlap with Anger-related expressions.

Table 3: Model Performance by Emotion Category

Emotion	F1-Score	Support
Love	0.7276	779
Joy	0.7872	1304
Anger	0.7194	1296
Surprise	0.8084	784
Sadness	0.7540	1350
Fear	0.8039	630
Hate	0.6460	721

4.9 Training Dynamics

Figures 2 and 3 illustrate the training behavior of the proposed model. The training loss decreases steadily across epochs, while the validation Macro F1-score increases and stabilizes, indicating effective learning without significant overfitting. The use of cosine learning-rate scheduling with warm-up and gradient clipping contributes to stable convergence. Early stopping further prevents performance degradation in later epochs.

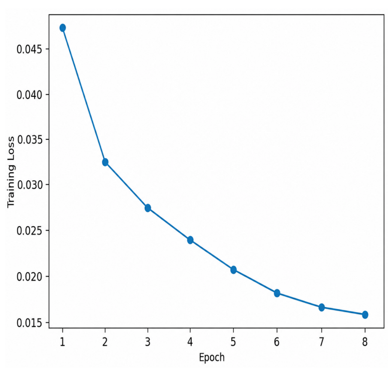

Figure 2: Training loss curve across epochs

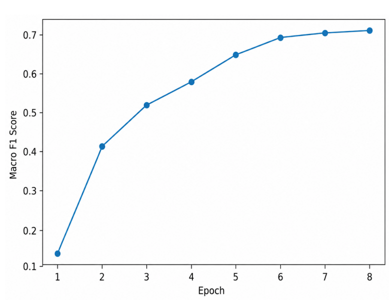

Figure 3: Validation Macro F1-score across epochs

4.10 Confusion and Error Analysis

Although the proposed XLMR-OPT framework achieves strong overall performance, several challenges remain in distinguishing semantically overlapping emotions. Most prediction errors occur between emotionally correlated categories such as Anger and Hate, as well as Joy and Love. These emotion pairs frequently share similar contextual and lexical patterns in Banglish social media discourse, making precise classification difficult.

The model also struggles with sarcastic, implicit, and context-dependent emotional expressions. In many cases, users express negative emotions indirectly through humor, irony, or informal slang, which reduces prediction reliability even for transformer-based architectures. Furthermore, inconsistent Romanized spelling and non-standard transliteration introduce significant lexical variability across samples.

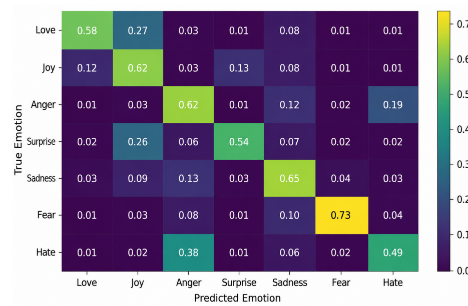

Figure 4: Frequently confused emotion pairs in the multi-label classification task

Table 4: Examples of Misclassified Samples

Text	True Label	Predicted Label
doctor ta bhalo but medicine ta expensive	None	Love
waittt amar gram er bari o shariatpur same place wahhhh	Love	None
samne asbe haguu variant	Anger	Joy
paltu app age valo celo	Anger	Joy
story is kind of gross but very interesting	Joy	None

The qualitative examples further demonstrate that the proposed model performs well on explicit emotional expressions but occasionally struggles with overlapping emotions and implicit semantic cues.

4.11 Label Correlation Analysis

Figure 5 illustrates the correlation structure among emotion labels in the BEmoMix dataset. Strong positive correlations are observed between Joy and Love, as well as Anger and Hate, reflecting the natural co-occurrence of related emotional categories in Banglish social media text.

Several emotion pairs also exhibit weak or negative correlations, confirming the inherently multi-label nature of emotion expression. These findings further justify the use of adaptive per-class threshold optimization for handling varying label distributions and overlapping emotional patterns.

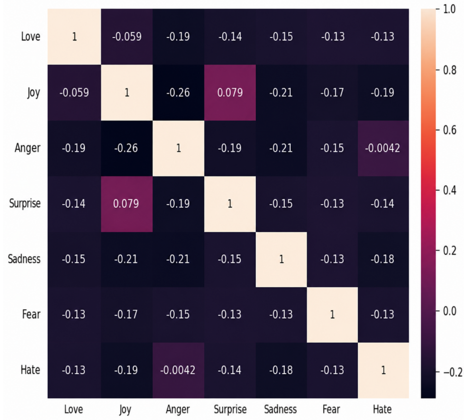

Figure 5: Label correlation matrix for emotion categories

4.12 Attention Analysis

Attention analysis reveals that the proposed transformer model assigns higher weights to emotionally salient tokens and contextual cues during prediction. Emotion-indicative words such as "happy", "scared", "khub rag", and "mon kharap" consistently receive stronger attention compared to neutral tokens, indicating that the self-attention mechanism effectively captures emotionally relevant information.

The model also demonstrates the ability to preserve contextual relationships across mixed Bangla-English segments, which is particularly important in Banglish social media text where emotional meaning often depends on code-switched expressions and contextual intensity modifiers. These observations suggest that transformer-based architectures learn meaningful semantic dependencies beyond simple keyword matching.

4.13 Hard Case Analysis

Table 5 presents several challenging examples where the proposed model fails to identify the target emotion labels correctly. These hard cases typically involve ambiguous phrasing, domain-specific slang, weak emotional indicators, or incomplete contextual information.

In multiple cases, the model incorrectly predicts Joy for sarcastic or negatively toned statements because of the absence of explicit emotional keywords. Some samples also contain highly implicit emotional intent, making semantic interpretation difficult even for advanced multilingual transformers.

These findings highlight the limitations of current contextual representation learning in low-resource code-mixed settings and suggest the need for richer semantic reasoning, sarcasm-aware modeling, and external knowledge integration in future work.

Table 5: Examples of Hard Misclassified Cases

Text	True Emotion	Predicted Emotion
price vai rog 7 thike onek besi dam s23 ultra	Anger	Joy
apnar shathe oi bideshi mohilati ke	Joy	Fear
samne asbe haguu variant	Anger	Joy
college er peon ke diye khata dekhano hocche boleo shona jacche	Hate	None
paltu app age valo celo	Anger	Joy

5 Discussion

This paper introduced **BEmoMix**, a large-scale multi-label emotion detection dataset containing 27,809 Banglish (Bangla- English code-mixed) social media texts annotated with seven emo- tion categories. The proposed XLMR-OPT framework achieved a Macro F1-score of 0.7495, establishing a strong baseline for Banglish emotion classification.

Unlike prior single-label studies, this work focuses on multi-label emotion detection in Romanized Banglish text, addressing challenges such as noisy transliteration, code-mixing, overlapping emotions, and imbalanced label distributions. The results demonstrate that multilingual transformer architectures effectively capture contextual emotional patterns in low-resource environments.

Overall, BEmoMix provides the first large-scale Banglish-specific multi-label emotion benchmark and creates opportunities for future research in label-dependency-aware learning, cross-lingual transfer, multimodal emotion understanding, fairness-aware NLP systems, and large language model fine-tuning.

5.1 Comparison with Prior Work

Table 6 compares the proposed XLMR-OPT framework with several existing Bangla and Banglish emotion and sentiment analysis studies. Direct comparison across studies should be interpreted cautiously due to differences in datasets, annotation schemes, language settings, label configurations, and evaluation protocols. Nevertheless, the comparison provides useful insight into the effectiveness of transformer-based approaches for multilingual and code-mixed emotion classification.

Earlier studies such as EmoNoBa primarily focused on fine-grained single-label emotion classification using hybrid N-gram and recurrent neural architectures, achieving a Macro F1-score of approximately 0.43 [9]. In contrast, BEmoLexBERT employed transformer-based multilingual representations for multi-label Bangla emotion classification and reported strong weighted F1 performance on monolingual Bangla text [11]. Similarly, BnSentMix addressed Bengali-English code-mixed sentiment analysis using transformer-based models under a sentiment polarity setting rather than fine-grained emotion classification [4].

Unlike prior approaches, the proposed BEmoMix framework specifically addresses the more challenging task of multi-label emotion detection in noisy Romanized Banglish text. The dataset contains substantial linguistic variability caused by informal transliteration, multilingual code-switching, overlapping emotional semantics, and uneven multi-label co-occurrence patterns. These characteristics make the task considerably more complex than conventional monolingual or single-label emotion classification settings.

Experimental results demonstrate that the proposed XLMR-OPT framework achieves a Macro F1-score of 0.7495 while maintaining strong class-wise balance across multiple emotion categories. The findings highlight the effectiveness of multilingual transformer architectures combined with hybrid contextual aggregation and adaptive threshold optimization for low-resource code-mixed NLP applications.

Table 6: Comparison with Existing Bangla/Banglish Emotion and Sentiment Studies

Study / Dataset	Model	Task / Labels	Reported Score
EmoNoBa [9]	Hybrid N-gram and BiLSTM models	6 emotions	Macro F1 ≈ 0.43
BEmoLexBERT [11]	BanglaBERT-based hybrid transformer	Multi-label Bangla emotions	Weighted F1 = 0.8273
BnSentMix [4]	Transformer-based models	4 sentiment labels	F1 ≈ 0.69
Bengali & Banglish Dataset [8]	Transformer-based benchmark	6 emotions	Weighted F1 ≈ 0.65
BEmoMix (Proposed)	XLMR-OPT	7 emotions (multi-label)	Macro F1 = 0.7495

The proposed task is substantially more challenging because BEmoMix focuses on multilingual code-mixed Banglish text with noisy Romanized transliteration and complex multi-label emotional overlap, whereas several prior studies are conducted on comparatively cleaner monolingual Bangla datasets. Consequently, direct numerical comparison across studies should be interpreted cautiously.

6 Conclusion

This paper introduced **BEmoMix**, a large-scale Banglish multi-label emotion detection dataset designed for code-mixed social media text. Experimental evaluation demonstrated that the proposed XLMR-OPT framework achieves a Macro F1-score of 0.7495, outperforming recurrent and embedding-based baselines in multilingual emotion classification tasks. The results further highlight the effectiveness of transformer-based contextual representation learning for handling noisy, transliterated, and code-mixed Banglish expressions.

Overall, BEmoMix provides a strong benchmark for multi-label Banglish emotion analysis and contributes toward advancing emotion-aware NLP research for low-resource multilingual communities.

7 Limitations and Future Work

Despite achieving strong overall performance, several limitations remain. The dataset primarily focuses on Romanized Banglish social media text and may not fully capture linguistic variations across domains and regional dialects. The proposed model also struggles with sarcasm, implicit sentiment, and overlapping emotional expressions in noisy code-mixed environments.

Another challenge arises from imbalanced emotion distributions, where minority labels receive comparatively fewer training examples. Although focal optimization and adaptive threshold calibration improve robustness, multi-label emotion detection in low-resource multilingual settings remains difficult. Future work may explore label-dependency-aware architectures, cross-lingual transfer learning [17], multimodal emotion understanding, and large language model fine-tuning for multilingual NLP applications [15].

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